Growsure

Enterprise Wealth Management & Insurance Platform

A full-stack, decoupled software application implementation featuring React & Redux client integration with interchangeable Java Spring Boot and .NET Core REST backends.

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Version: 1.0.0 (Production-Ready)

1. System Architecture Overview

Growsure is engineered using a decoupled, dual-backend enterprise architecture. The React SPA client manages dynamic routing, state propagation, and authentication. Developers or administrators can alternate target REST backends (Spring Boot on Port 8080 or ASP.NET Core on Port 8081) directly through an API swapper mechanism in the user interface. Both servers implement identical service specifications and map to a shared MySQL database, maintaining complete transparent state synchronization.

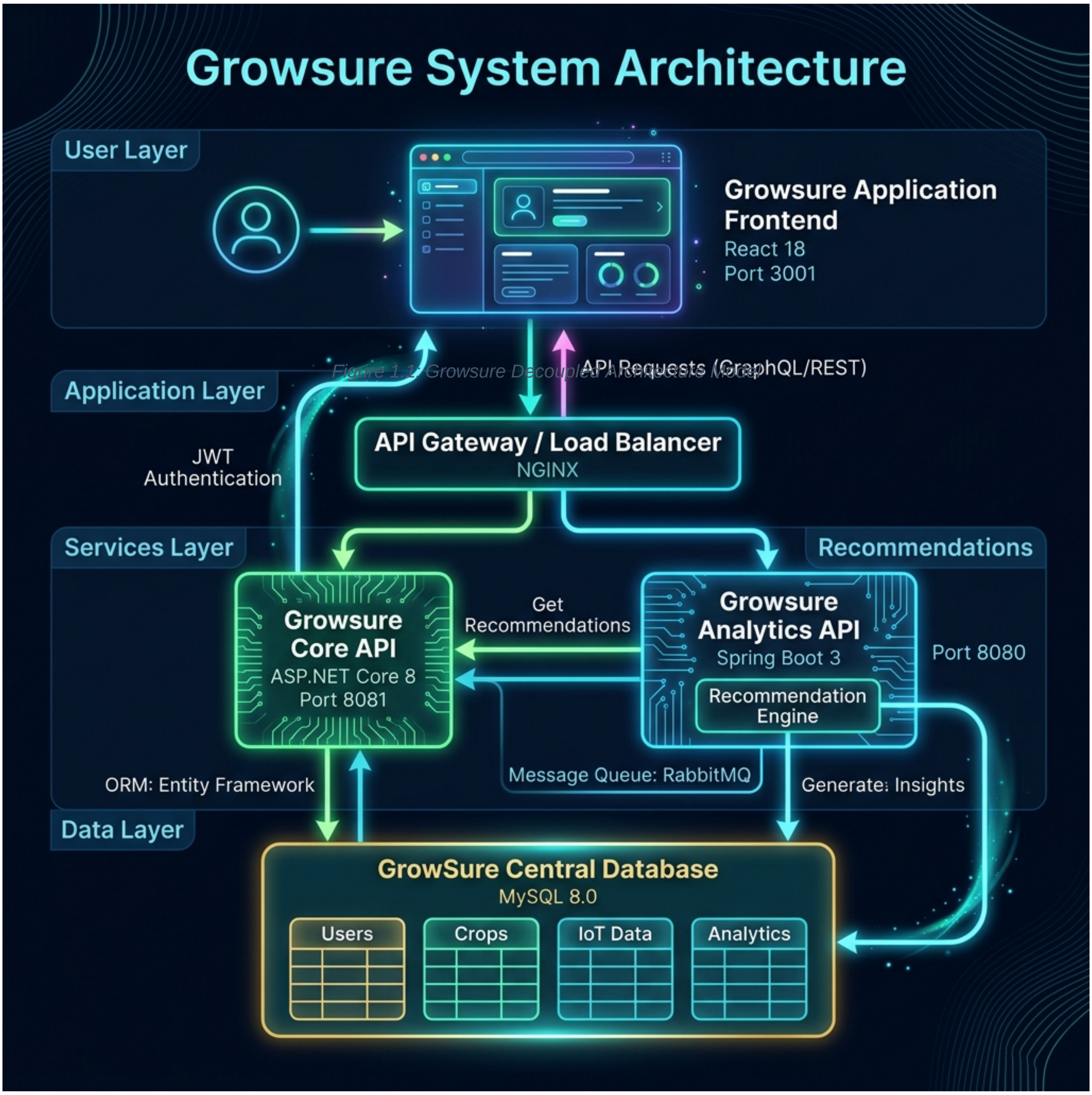

Figure 1.1: Growsure Decoupled Architecture Model

2. Database Schema & Relational Models

The shared MySQL database contains 12 core tables mapping user profiles, insurers, policy holders, policies, transactions, and mutual funds. Table schemas are strictly aligned across Java JPA annotations and C# Entity Framework Core configurations.

Table Name	Key Fields	Description & Relationships
users	*user_id (PK)*	Credentials and roles (ADMIN, INSURER, POLICY_HOLDER).
policy_holders	*id (PK), user_id (FK)*	Demographic profiles linked 1-to-1 with users.
insurers	*id (PK), user_id (FK)*	Corporate partners with IRDAI licensing state indicators.
policies	*policy_id (PK), insurer_id (FK)*	Insurance product offerings (HEALTH, LIFE, MOTOR, TRAVEL).
purchased_policies	*purchase_id (PK), policy_id (FK)*	Active contracts generated upon successful payments.
claims	*claim_id (PK), purchase_id (FK)*	Claim invoices audited by the automated AI engine.
funds	*fund_id (PK)*	Mutual funds including risk parameters, CAGR history, and AUM size.
investments	*investment_id (PK), fund_id (FK)*	SIP allocations and lumpsum holdings of policy holders.

3. AI Financial Recommendation Engine

Growsure implements a comprehensive wealth advisor system. It suggests custom life/health insurance coverage limits and optimizes mutual fund asset allocation depending on user demographics. Recommendations run in two modes: Real AI Mode (querying OpenAI GPT-4o) and Mock Rule-Based Mode (applying local fallback algorithms).

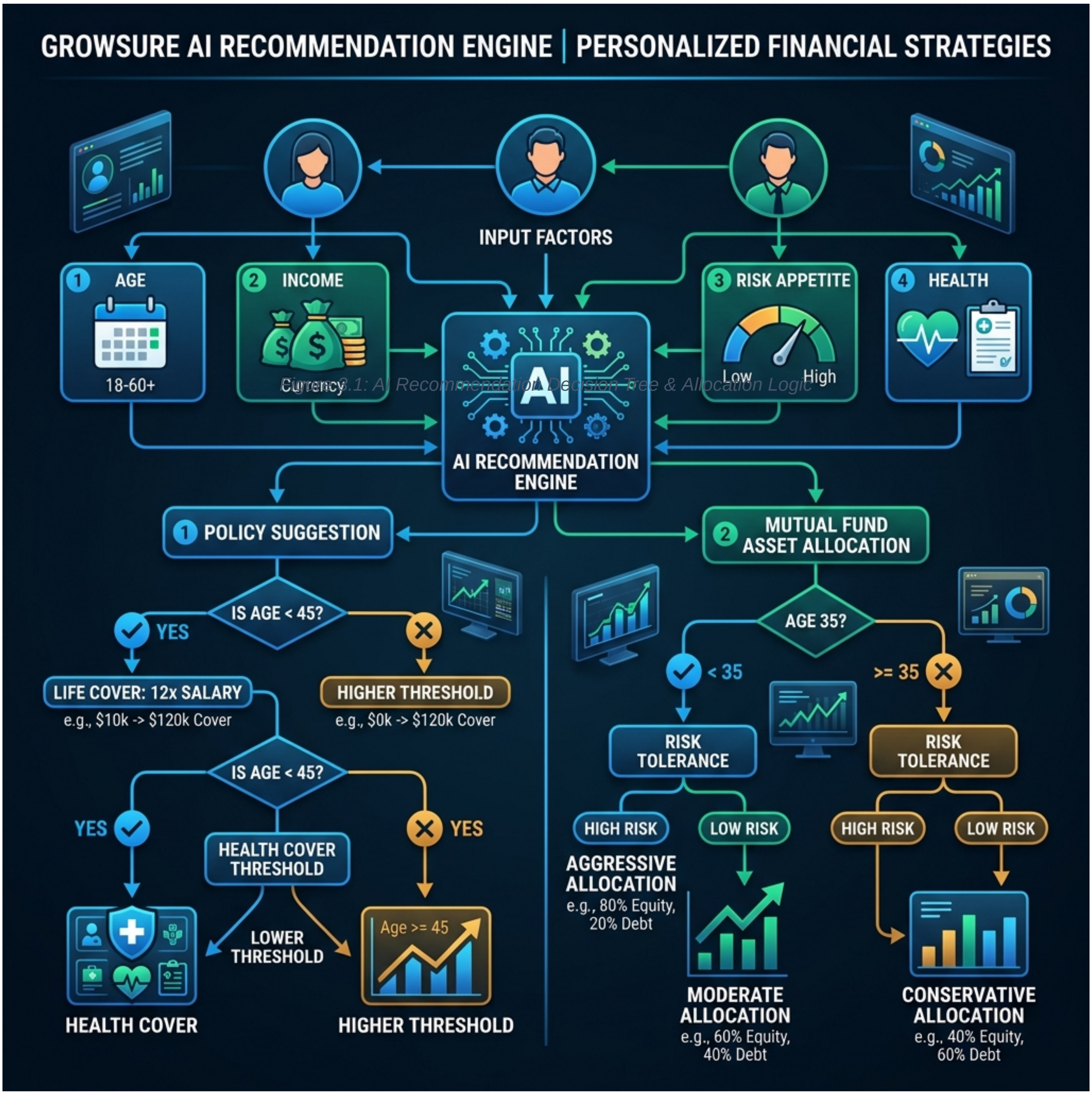

Figure 3.1: AI Recommendation Decision Tree & Allocation Logic

3.1 Insurance Policy Suggestion Logic

The insurance recommendation module calculates optimal coverage limits to protect the user's family and assets without generating unnecessary premium loads.

A. Term Life Insurance Sizing

The system sizes Term Life Coverage based on the Human Life Value (HLV) methodology. The sum insured is set to exactly 12 times the customer's annual salary to replace their income in the event of unforeseen loss:

Suggested Life Coverage = Annual Income * 12

B. Health Insurance Sizing

Health policy sizes adapt to age-based risk limits and chronic illness variables. The underwriting rules are defined as follows:

Health Cover Rules:

- If Age > 45 OR Health Condition contains "diabetes" or "heart":
  Suggested Sum Insured = INR 1,000,000 (10 Lakhs)
- If Age <= 45 AND Health Condition is Good:
  Suggested Sum Insured = INR 500,000 (5 Lakhs)

3.2 Mutual Fund Asset Allocation

Growsure constructs custom asset portfolios based on the user's age and risk appetite to capture compounding capital growth.

Aggressive Portfolio (Age < 35 OR Risk = "HIGH")

Fund Name	Asset Class	Allocation	CAGR
Quant Small Cap Fund	Small Cap Equity	**40%**	34.5%
Nippon India Small Cap Fund	Small Cap Equity	**30%**	29.8%
HDFC Mid-Cap Opportunities Fund	Mid Cap Equity	**30%**	23.2%

Conservative Portfolio (Age >= 35 AND Risk != "HIGH")

Fund Name	Asset Class	Allocation	CAGR

Parag Parikh Flexi Cap Fund

Flexi Cap / Hybrid

50%

21.4%

Mirae Asset Large Cap Fund

Large Cap Equity

30%

15.6%

SBI Magnum Gilt Fund

Sovereign Debt

20%

7.8%

4. Core Business Workflows

4.1 Smart Claim Fraud Auditing

When claims are filed, Growsure automatically triggers an AI claims auditor. The baseline claim fraud probability is set at 15.0%. If the requested claim amount exceeds INR 100,000, the system increases the fraud score to 38.0% and flags the record for manual validation. If the files or incident description match duplicate tag profiles (indicating double invoice submissions), the score is set to 78.0%, recommending rejection and triggering a high-priority system audit log entry.

4.2 Client Interceptor & Port Swapper

The React application intercepts all outgoing HTTP requests using a dynamic Axios wrapper. This fetches the targeted backend URL from local storage, shifting ports dynamically. Since both servers share matching symmetric security signatures, target backend swops execute seamlessly without logging the user out.